



Parameter-efficient multimodal adaptation of vision foundation models for remote sensing semantic segmentation

Haocheng Li^a, Juepeng Zheng^{b,c,*}, Shuangxi Miao^a, Ruibo Lu^d, Guosheng Cai^d, Haohuan Fu^{c,g}, Jianxi Huang^{a,e,f}^{**}

^a College of Land Science and Technology, China Agricultural University, Beijing, China

^b School of Artificial Intelligence, Sun Yat-Sen University, Zhuhai, China

^c National Supercomputing Center in Shenzhen, Shenzhen, China

^d School of Surveying and Land Information Engineering, Henan Polytechnic University, Jiaozuo, China

^e Faculty of Geosciences and Engineering, Southwest Jiaotong University, Chengdu, China

^f School of Electronic Engineering, Guangxi University of Science and Technology, Liuzhou, China

^g Tsinghua Shenzhen International Graduate School, Tsinghua University, Shenzhen, China

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ABSTRACT

Multimodal remote sensing semantic segmentation improves scene understanding by integrating complementary information from heterogeneous Earth observation data. Although pretrained Vision Foundation Models (VFMs) provide strong general-purpose representations, adapting them to multimodal tasks typically requires updating large modality-specific encoders, and optimization tends to be dominated by the optical modality, leaving auxiliary cues under-utilized. To address these challenges, we propose MoBaNet, a parameter-efficient multimodal adaptation framework that keeps the VFM backbone frozen and concentrates trainable capacity on three complementary components. First, the Cross-modal Prompt-Injected Adapter (CPIA) generates semantic prompts from paired modalities and injects them into lightweight bottleneck adapters to enable cross-modal interaction under the frozen backbone. Second, the Difference-Guided Gated Fusion Module (DGFM) exploits cross-modal discrepancy to guide channel-wise and spatially adaptive feature fusion. Third, the training-only Modality-Conditional Random Masking (MCRM) strategy locally corrupts one modality at a time and applies hard-pixel auxiliary supervision to modality-specific branches, encouraging more robust use of complementary cues. Extensive experiments on the ISPRS Vaihingen and Potsdam benchmarks show that MoBaNet achieves the best overall performance among the compared methods, while its default DINOv2-ViT-B configuration uses only 5.43% trainable parameters and remains robust under partial modality degradation.

1. Introduction

Remote sensing semantic segmentation aims to assign a semantic label to each pixel in high-resolution Earth observation (EO) imagery (Zhu et al., 2017), serving as a fundamental building block for a wide range of applications, including land-cover mapping, urban monitoring, precision agriculture, and disaster assessment (Dakogiannis et al., 2020; Chen et al., 2021; Turkoglu et al., 2021; Zheng et al., 2021). Compared with semantic segmentation in natural scenes, RS imagery typically exhibits larger scale variations, more frequent occlusions, and more complex backgrounds (Maggiori et al., 2017; Volpi and Tuia, 2017; Liu et al., 2018). Moreover, EO data are highly susceptible to illumination changes, cast shadows, seasonal variations, sensor-induced imaging artifacts, and domain shifts across

acquisition conditions (Zhang et al., 2014; Bai et al., 2022; Ma et al., 2024b). These factors often lead to pronounced appearance ambiguity and inter-class confusion, posing more stringent requirements on the robustness of dense prediction models (Ma et al., 2019). To mitigate these challenges, researchers have increasingly incorporated complementary sensing modalities beyond RGB (Schmitt and Zhu, 2016; J. Li et al., 2022). By fusing the physical cues encoded in heterogeneous modalities, multimodal approaches can compensate for the observational limitations of optical imagery alone in complex environments, thereby improving the completeness and stability of scene interpretation (Audebert et al., 2018; Yang et al., 2021).

Although multimodal data provide rich complementary cues for RS semantic segmentation, it remains challenging to fuse heterogeneous

* Corresponding author at: School of Artificial Intelligence, Sun Yat-Sen University, Zhuhai, China.

** Corresponding author at: College of Land Science and Technology, China Agricultural University, Beijing, China.

E-mail addresses: haoc.li@cau.edu.cn (H. Li), zhengjp8@mail.sysu.edu.cn (J. Zheng), jxhuang@swjtu.edu.cn (J. Huang).

modalities efficiently and reliably within a unified framework (Parmehr et al., 2014; Sun and Wang, 2018). On the one hand, different modalities exhibit substantial discrepancies in imaging mechanisms, numerical distributions, and noise characteristics — often exacerbated by resolution mismatch and registration errors — which complicate cross-modal feature alignment and interaction modeling (Zheng et al., 2022). On the other hand, existing fusion strategies may be dominated by optical information during optimization or discard beneficial modality-specific information, such that complementary cues cannot be consistently translated into performance gains (Zheng et al., 2025; S. Wei et al., 2025; T. Wei et al., 2025). In recent years, Vision Transformers (ViTs) (Dosovitskiy et al., 2021) and vision foundation models (VFM) (Oquab et al., 2024; Kirillov et al., 2023) pretrained on large-scale data have demonstrated stronger general-purpose representations, offering new opportunities for RS semantic segmentation. However, in multimodal settings, mainstream approaches typically instantiate separate backbones for different modalities and perform full fine-tuning (Wang et al., 2022; X. Li et al., 2022), which incurs substantial training and storage overhead. More importantly, such designs still struggle to prevent optimization from converging to optical-dominant shortcut solutions, suggesting that architectural symmetry does not necessarily yield balanced contributions across modalities (Zheng et al., 2025; S. Wei et al., 2025; Niu et al., 2022).

To address the above issues, we propose MoBaNet, a parameter-efficient multimodal adaptation framework for remote sensing semantic segmentation. Rather than fully fine-tuning large modality-specific encoders, MoBaNet follows a frozen-backbone adaptation principle: it preserves the pretrained representation space of VFMs and concentrates the limited trainable capacity on multimodal-specific operations. Specifically, lightweight modules are introduced to handle modality-specific tokenization, deep cross-modal interaction, discrepancy-aware feature fusion, and training-time robustness regularization. This design is formulated for optical-auxiliary multimodal EO inputs and is not tied to DSM-specific operators; in this work, we instantiate and evaluate it on two representative optical-DSM urban segmentation benchmarks.

Specifically, to overcome the limited deep cross-modal interaction caused by freezing the backbone, we first design a Cross-modal Prompt-Injected Adapter (CPIA). Without disrupting the pretrained feature space of the foundation model, CPIA generates shared semantic prompts from both modalities and injects them into the bottleneck of lightweight adapters, enabling implicit semantic-level interaction and modulation across modalities (Houlsby et al., 2019). Second, to obtain compact and discriminative multimodal representations for decoding, we propose a Difference-Guided Gated Fusion Module (DGFM). DGFM explicitly exploits cross-modal discrepancy to perform channel-wise and spatially adaptive fusion on paired stage features, thereby enhancing complementary information integration in a lightweight manner. This design not only suppresses the interference of noisy or inconsistent responses during fusion, but also yields more robust multimodal representations with low computational overhead. Finally, to alleviate the tendency of multimodal models to exhibit optical-dominant bias during training, we introduce a Modality-Conditional Random Masking (MCRM) strategy. MCRM randomly applies local masking to only one modality at the batch level and imposes hard-pixel auxiliary supervision on modality-specific branches under partial modality degradation. This mechanism encourages the model to exploit informative cues from the complementary modality when the dominant modality becomes unreliable, and thus improves the robustness and balance of cross-modal representation learning. The main contributions of this paper are as follows:

- We propose MoBaNet, a parameter-efficient framework that adapts pretrained Vision Foundation Models to multimodal remote sensing semantic segmentation by freezing the backbone and reallocating a small trainable budget to multimodal-specific modules.
- We design two complementary modules under the frozen backbone: the CPIA injects shared prompts into lightweight bottleneck adapters before each stage to enable deep cross-modal interaction, and the DGFM leverages cross-modal discrepancy to perform adaptive gated fusion after each stage.
- We introduce MCRM, a training-only strategy that combines modality-conditional masking with hard-pixel auxiliary supervision to mitigate over-reliance on a single dominant modality.
- Experiments on two ISPRS multimodal benchmarks show consistent accuracy gains and improved robustness, while the default DINOv2-ViT-B configuration trains only 5.43% of its parameters.

2. Related work

2.1. Multimodal fusion for remote sensing segmentation

Multimodal remote sensing semantic segmentation improves class separability and robustness by fusing complementary sensing cues, e.g., multispectral appearance, DSM-derived elevation, and SAR scattering characteristics (Zhu et al., 2017; J. Li et al., 2022). Existing methods typically adopt dual- or multi-branch encoders to extract modality-specific features, and can be broadly grouped into three paradigms according to the fusion stage: *early fusion* at the input level, *late fusion* at the prediction/decision level, and *feature-level fusion* within the encoder-decoder pipeline (Baltrušaitis et al., 2019). Among them, feature-level fusion has become a predominant direction because it enables explicit cross-modal interaction and semantic alignment while preserving modality-specific characteristics (J. Li et al., 2022; Liu et al., 2024a).

Building upon feature-level fusion, recent research has evolved from parallel architectures toward interaction-enhanced designs, aimed at boosting the effective contribution of auxiliary modalities while suppressing redundant information (J. Li et al., 2022). These methodologies can be broadly categorized into three mechanisms: (i) *Cross-modal guidance*, which constrains the decoding process of one modality using structural or semantic cues from another (e.g., Gather-to-Guide (Zheng et al., 2022) ensures consistency between auxiliary cues and semantic decoding); (ii) *Selective fusion*, utilizing gating mechanisms or attention weights to attenuate modality-specific noise and highlight complementary regions (Hosseinpour et al., 2022), as exemplified by CEGFNet's (Zhou et al., 2022) explicit gating for redundancy reduction; and (iii) *Intra-/inter-modality joint interaction*, which establishes explicit cross-modal interaction pathways while maintaining modality-specific representations. A representative recent work is MFNet (Ma et al., 2025), which adapts SAM to multimodal remote sensing semantic segmentation through a multimodal fine-tuning framework and a deep fusion module.

Meanwhile, the emergence of datasets and benchmarks tailored for heterogeneous modalities has also facilitated the development of multimodal remote sensing segmentation methods (Schmitt and Zhu, 2016; Liu et al., 2024b). For example, WHU-OPT-SAR and its accompanying MCANet (X. Li et al., 2022) further validated the complementary benefits of heterogeneous modalities through cross-modal attention and multi-level feature fusion. Toward stronger interaction modeling, Transformer-based fusion has gradually become an important direction. FTransUNet (Ma et al., 2024c) introduces a fusion Transformer to perform global interaction over high-level semantics on top of a CNN backbone that captures shallow details, whereas TMFNet (Liu et al., 2024a) and related approaches further explore DSM-oriented attention interactions and correlation modeling to enhance long-range dependency modeling and cross-modal semantic coupling. Overall, existing multimodal RS segmentation methods have progressively evolved from fusion-level design to stronger interaction modeling, with increasing emphasis on selective noise suppression/alignment and global context modeling (Li et al., 2026).

2.2. Foundation models and parameter-efficient adaptation

Unlike traditional CNN-based architectures, VFMs are largely built upon ViTs and learn robust visual representations via self-supervised or weakly supervised pre-training. Among them, SAM (Kirillov et al., 2023) is centered on *promptable segmentation*, encapsulating segmentation as a prompt-driven universal interface and showing competitive cross-domain transferability. Recent studies have further introduced SAM into remote sensing semantic segmentation by incorporating object- and boundary-aware constraints to improve dense prediction quality (Ma et al., 2024a). DINOv2 (Oquab et al., 2024), in contrast, follows a self-supervised representation learning paradigm, acquiring robust patch- and image-level features and serving as a general-purpose backbone for downstream tasks such as classification, detection, and segmentation. Meanwhile, the remote sensing community has also begun exploring foundation models tailored for multimodal data, such as RingMo-SAM (Yan et al., 2023) with a dedicated prompt encoder and class-decoupled mask decoder, and SkySense (Guo et al., 2024) with multimodal spatiotemporal encoding for joint cross-modal modeling. Overall, foundation models offer strong transferability and data-efficient adaptation for multimodal RS understanding.

However, the parameter scale and training cost of foundation models substantially raise the barrier to full fine-tuning. This has stimulated growing interest in PEFT. The central idea of PEFT is to keep the backbone frozen and introduce only a small set of trainable modules for task- or domain-specific adaptation. Representative strategies include Adapters (Houlsby et al., 2019), which insert lightweight bottleneck modules between layers; LoRA (Hu et al., 2022), which injects low-rank trainable updates into linear projections; and visual prompt tuning (VPT) (Jia et al., 2022), which learns a small number of prompt vectors in the input token space. Prior studies have systematically reviewed PEFT methods and broadly categorized them into additive, reparameterization-based, and prompt-based schemes. Overall, PEFT can achieve competitive performance under a limited trainable-parameter budget while substantially reducing training and storage overhead. In remote sensing, PEFT is often used with foundation models to bridge the gap between general-purpose visual representations and domain-specific texture statistics and semantic requirements (Hu et al., 2024b,a; Cao et al., 2026).

Although recent VFM adaptation studies have explored frequency-spatial dual-domain fine-tuning for domain generalized remote sensing semantic segmentation (Shen et al., 2026), most existing PEFT methods remain primarily designed for single-modality vision tasks, and directly transferring them to multimodal remote sensing scenarios still faces critical bottlenecks (Hu et al., 2024b; Zou et al., 2025). First, deep cross-modal interaction is insufficient: in mainstream dual-stream architectures, if adapters are configured independently for each modality, the frozen backbone branches lack adequate channels for deep information exchange. As a result, heterogeneous modalities are processed in a manner closer to parallel feature extraction with weakly coupled fusion, making it difficult to achieve deep complementary coupling during encoding (Ma et al., 2024c). Second, modality imbalance may be reinforced under PEFT: frozen foundation models often inherit a preference for optical cues, and standard PEFT does not explicitly model the competition between modalities (Wang et al., 2023; Chen et al., 2024). Consequently, the limited trainable parameters may still favor optical-dominant shortcut solutions, further diminishing the effective contribution of the auxiliary modality (Gao et al., 2024). These limitations suggest that frozen-backbone adaptation and multimodal fusion should be considered jointly rather than treated as independent steps. Accordingly, MoBaNet reallocates the limited trainable capacity to cross-modal prompt interaction, discrepancy-guided fusion, and training-time modality-robust regularization under a frozen VFM backbone (see Fig. 1).

3. Method

In this section, we first introduce the overall framework in Section 3.1. The details of CPIA, DGFM, and MCRM are presented in Sections 3.2, 3.3, and 3.4, respectively.

3.1. Overall architecture

MoBaNet is designed as a parameter-efficient dual-stream adaptation framework for multimodal remote sensing segmentation, coupling frozen-backbone feature extraction with lightweight cross-modal interaction and fusion.

Given an input multimodal image pair consisting of a selected three-band optical image $\mathbf{I}_{\text{opt}} \in \mathbb{R}^{B \times 3 \times H \times W}$ and an auxiliary modality $\mathbf{I}_{\text{aux}} \in \mathbb{R}^{B \times C_a \times H \times W}$, our goal is to predict a pixel-wise class probability map $\mathbf{P} \in \mathbb{R}^{B \times K \times H \times W}$. Here, $\mathbf{I}_{\text{opt}}$ denotes the dataset-specific three-band optical input used in our experiments, rather than necessarily RGB channels. As detailed in Section 4.1, Vaihingen uses NIR-R-G imagery, whereas Potsdam uses RGB imagery.

We adopt a pretrained ViT as the feature extraction backbone and keep its parameters Θ_{backbone} frozen during training. To accommodate the statistical distribution shift and channel mismatch of heterogeneous auxiliary modalities, we design an asymmetric patch embedding scheme: the optical branch reuses and freezes the pretrained three-channel patch embedding E_{opt} , whereas the auxiliary branch introduces an additional embedding E_{aux} whose parameters are trainable. Since the selected optical input is kept as a three-channel tensor for both datasets, it is directly compatible with E_{opt} without additional channel projection or modification of the pretrained patch-embedding weights. After patch embedding, both modalities are represented as a 2-D token grid:

$$\mathbf{X}_0 = E_{\text{opt}}(\mathbf{I}_{\text{opt}}), \quad \mathbf{Y}_0 = E_{\text{aux}}(\mathbf{I}_{\text{aux}}), \quad (1)$$

$$\mathbf{X}_0, \mathbf{Y}_0 \in \mathbb{R}^{B \times H' \times W' \times C}, \quad H' = \frac{H}{p}, \quad W' = \frac{W}{p}. \quad (2)$$

Here, p denotes the patch size and C is the embedding dimension. We apply a shared and frozen absolute positional encoding $\mathbf{P}_{\text{abs}}$ to both streams. To maintain consistent spatial priors under varying input resolutions, $\mathbf{P}_{\text{abs}}$ is resized via bicubic interpolation and then added to the token embeddings.

The feature extraction stage consists of L stacked Transformer blocks. Following the multi-level feature tapping configuration, we divide the encoder into S stages according to the selected tap indices $\{t_s\}_{s=1}^S$, where $S = 4$ by default. The s th stage is defined as the sequence of Transformer blocks from block $t_{s-1} + 1$ to block t_s , with $t_0 = 0$. Within each stage, the two modalities are structurally aligned to share the same unimodal modeling capacity and optimization path, while cross-modal interaction is introduced only through a small set of trainable PEFT modules. Given the input tokens of the l th block, $\mathbf{X}_{l-1}, \mathbf{Y}_{l-1} \in \mathbb{R}^{B \times H' \times W' \times C}$, the shared Pre-LN unimodal modeling is formulated as

$$\tilde{\mathbf{X}}_l = \mathbf{X}_{l-1} + \text{SA}(\text{LN}(\mathbf{X}_{l-1})), \quad \tilde{\mathbf{Y}}_l = \mathbf{Y}_{l-1} + \text{SA}(\text{LN}(\mathbf{Y}_{l-1})), \quad (3)$$

$$\tilde{\mathbf{X}}_l = \tilde{\mathbf{X}}_l + \text{MLP}(\text{LN}(\tilde{\mathbf{X}}_l)), \quad \tilde{\mathbf{Y}}_l = \tilde{\mathbf{Y}}_l + \text{MLP}(\text{LN}(\tilde{\mathbf{Y}}_l)). \quad (4)$$

The parameters of $\text{SA}(\cdot)$ and $\text{MLP}(\cdot)$ are shared across the two streams (shared weights) and are frozen together with the backbone.

Rather than inserting trainable modules into every Transformer block, we adopt a stage-wise insertion strategy. For each selected stage, we place CPIA before the stage encoder and DGFM after the stage encoder:

- **CPIA (implicit semantic modulation):** before the input tokens enter the s th stage, CPIA generates shared semantic prompts from the bimodal tokens and injects them into both streams via prompt-injected adapters, enabling cross-modal semantic modulation under a frozen backbone.

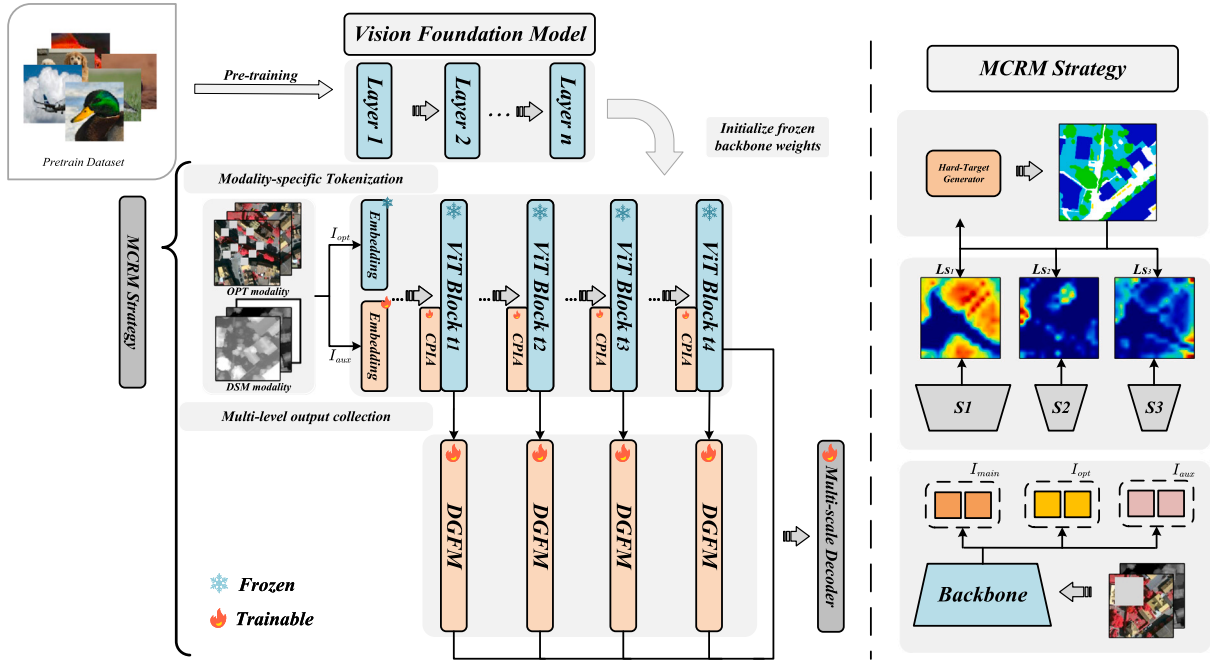


Fig. 1. Overall framework of the proposed multimodal segmentation model with MCRM training. A pretrained vision foundation model initializes a shared frozen ViT backbone. Optical and auxiliary inputs are first tokenized by modality-specific embeddings, then processed by a stage-wise dual-stream encoder, where CPIA is inserted before each selected stage and DGFM is applied after each selected stage to produce multi-level fused features for the decoder. During training, MCRM performs modality-conditional random masking and hard-pixel auxiliary supervision on one main branch (S1) and two auxiliary modality branches (S2, S3), while only the main branch is used for inference.

- **DGFM (explicit stage-level fusion):** after the s th stage produces its output features, DGFM adaptively fuses the paired modality features by leveraging their discrepancy cue to guide channel-wise and spatially adaptive fusion, yielding a compact multimodal representation for decoding.

Denote by $\mathbf{X}^s$ and $\mathbf{Y}^s$ the optical and auxiliary features output from the s th selected stage, respectively. DGFM produces the fused feature $\mathbf{F}^s$ for each stage. The multi-level fused features $\{\mathbf{F}^s\}_{s=1}^S$ are then fed into the decoder head to produce the main prediction $\mathbf{P}$. During training, we additionally attach two lightweight modality-specific auxiliary heads to generate P_{opt} and P_{aux} for MCRM supervision, while only the main fused branch is retained during inference. During training, the VFM backbone parameters Θ_{backbone} are kept frozen, while the lightweight trainable components, including the auxiliary patch embedding, CPIA, DGFM, decoder head, and training-time auxiliary heads, are optimized.

3.2. Cross-modal prompt-injected adapter

Under the frozen-backbone setting, relying solely on independent Transformer computations in the two streams leads to insufficient deep cross-modal semantic interaction. To address this issue, we propose the CPIA, which is inserted before each selected stage encoder and operates on the paired modality tokens. It introduces a lightweight mechanism based on *prompt generation* and *prompt-injected adapters*, enabling implicit semantic complementarity across modalities without disrupting the pretrained feature space. Specifically, CPIA consists of two components: a Cross-modal Prompt Generator (CPG), which produces modality-specific semantic prompts from the paired bimodal tokens, and a Prompt-Injected Adapter, which injects these prompts as a dynamic bias into the bottleneck of a lightweight adapter inserted into each stream. In the following, we describe the two components in turn.

Cross-modal prompt generator. Given the paired modality tokens before entering the s th stage, $\mathbf{X}_{\text{in}}^s$ and $\mathbf{Y}_{\text{in}}^s \in \mathbb{R}^{B \times H' \times W' \times C}$, CPG first projects each modality to a reduced dimension $C_p = \lfloor r_p C \rfloor$. The two projected features are then concatenated along the channel dimension, fused, and finally projected back to C to obtain a shared semantic base $\mathbf{Z}^s$:

$$\mathbf{X}^{\downarrow, s} = \mathbf{X}_{\text{in}}^s \mathbf{W}_d^{\text{opt}}, \quad \mathbf{Y}^{\downarrow, s} = \mathbf{Y}_{\text{in}}^s \mathbf{W}_d^{\text{aux}}, \quad (5)$$

$$\mathbf{Z}^s = \text{Up}(\text{Fuse}([\mathbf{X}^{\downarrow, s}; \mathbf{Y}^{\downarrow, s}])), \quad (6)$$

Here, $\text{Fuse}(\cdot)$ and $\text{Up}(\cdot)$ denote linear mappings. To mitigate the distribution shift between different modalities, we introduce a Task Feature Transformation (TFT) module that generates modality-specific prompts via channel-wise affine modulation. Specifically, we define

$$\mathbf{P}_{\text{opt}}^s = \mathbf{Z}^s + (\mathbf{Z}^s \odot \gamma_{\text{opt}} + \beta_{\text{opt}}), \quad (7)$$

$$\mathbf{P}_{\text{aux}}^s = \mathbf{Z}^s + (\mathbf{Z}^s \odot \gamma_{\text{aux}} + \beta_{\text{aux}}), \quad (8)$$

where $\gamma_m, \beta_m \in \mathbb{R}^C$ are learnable channel-wise parameters for modality $m \in \{\text{opt}, \text{aux}\}$ and are broadcast along the spatial dimensions $H' \times W'$.

Prompt-injected adapter. Conventional adapters are designed for uni-modal features. In CPIA, we inject the prompt vectors generated by CPG as a dynamic bias into the bottleneck of an adapter. Specifically, we insert a prompt-injected adapter with a bottleneck structure into each stream, and use the prompt as an additive modulation term on the reduced intermediate representation. For modality $m \in \{\text{opt}, \text{aux}\}$, the adapter mapping is defined as

$$\text{Adapter}(\mathbf{X}, \mathbf{P}) = \mathbf{X} + \sigma(\mathbf{X} \mathbf{W}_{\text{down}} + \mathbf{P} \mathbf{W}_{\text{prompt}}) \mathbf{W}_{\text{up}}, \quad (9)$$

where $\mathbf{W}_{\text{down}} \in \mathbb{R}^{C \times d}$, $\mathbf{W}_{\text{prompt}} \in \mathbb{R}^{C \times d}$, and $\mathbf{W}_{\text{up}} \in \mathbb{R}^{d \times C}$ are learnable projection matrices. The bottleneck dimension is $d = \lfloor r_a C \rfloor$, and $\sigma(\cdot)$ denotes the ReLU activation. We further apply dropout to improve generalization. The CPIA outputs are then given by

$$\mathbf{X}_{\text{cpia}}^s = \text{Adapter}(\mathbf{X}_{\text{in}}^s, \mathbf{P}_{\text{opt}}^s), \quad (10)$$

$$\mathbf{Y}_{\text{cpia}}^s = \text{Adapter}(\mathbf{Y}_{\text{in}}^s, \mathbf{P}_{\text{aux}}^s). \quad (11)$$

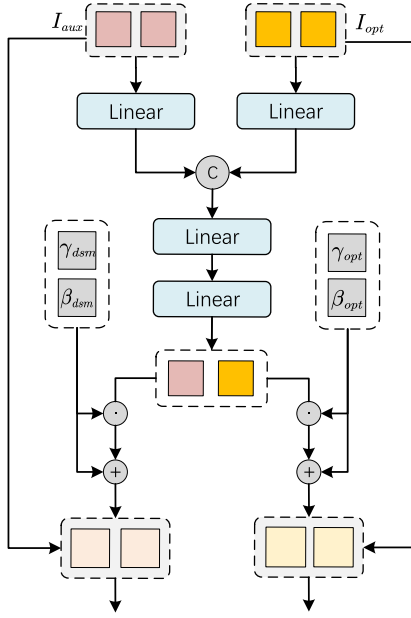


Fig. 2. Structure of the CPIA module. A shared semantic base is generated from paired optical and auxiliary tokens, transformed into modality-specific prompts by TFT, and injected into lightweight bottleneck adapters for symmetric cross-modal semantic modulation before each selected stage.

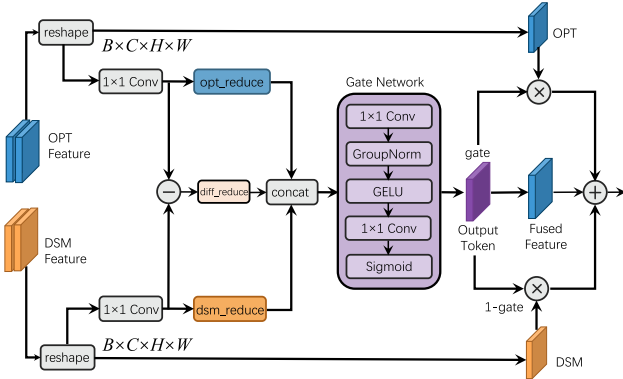


Fig. 3. Structure of the DGFM module. Reduced optical and DSM features, together with their discrepancy cue, are fed into a lightweight gate network to predict an adaptive gating map, which dynamically fuses the two modalities into a compact multimodal representation.

The modulated tokens $\mathbf{X}_{cpia}^s$ and $\mathbf{Y}_{cpia}^s$ are then fed into the s th stage encoder. This design symmetrically injects cross-modal prompts into the bottlenecks of both streams, thereby establishing a stable semantic communication channel under the frozen-backbone constraint (see Fig. 2).

3.3. Difference-guided gated fusion module

To obtain compact and discriminative multimodal representations for decoding, we introduce a lightweight *Difference-Guided Gated Fusion Module* (DGFM), as illustrated in Fig. 3. For each selected stage, DGFM is applied after the stage encoder to fuse the paired optical and auxiliary features. Unlike more complicated fusion designs that introduce dedicated refinement branches, DGFM performs adaptive fusion in a simple yet effective manner by exploiting modality-specific responses together with their explicit discrepancy cue.

Compact projection and discrepancy cue. Given the paired features from the s th selected encoder stage, $\mathbf{X}^s, \mathbf{Y}^s \in \mathbb{R}^{B \times L_s \times C_s}$ with $L_s = H_s W_s$, we first reshape them into 2-D feature maps, denoted by $\tilde{\mathbf{X}}^s, \tilde{\mathbf{Y}}^s \in \mathbb{R}^{B \times C_s \times H_s \times W_s}$. To reduce the computational cost of fusion, both modality features are projected into a compact channel space through 1×1 convolutions:

$$\mathbf{R}_x^s = \phi_x^s(\tilde{\mathbf{X}}^s), \quad \mathbf{R}_y^s = \phi_y^s(\tilde{\mathbf{Y}}^s), \quad (12)$$

where $\phi_x^s(\cdot)$ and $\phi_y^s(\cdot)$ denote learnable channel reduction mappings. The reduced features satisfy $\mathbf{R}_x^s, \mathbf{R}_y^s \in \mathbb{R}^{B \times C'_s \times H_s \times W_s}$, where $C'_s < C_s$.

Based on the reduced features, we explicitly construct a discrepancy cue:

$$\mathbf{D}^s = |\mathbf{R}_x^s - \mathbf{R}_y^s|, \quad (13)$$

where $\mathbf{D}^s \in \mathbb{R}^{B \times C'_s \times H_s \times W_s}$ captures cross-modal disagreement and provides direct evidence for adaptive fusion.

Difference-guided adaptive fusion. The reduced modality features and their discrepancy cue are concatenated to predict a channel-wise and spatially adaptive gating map:

$$\mathbf{G}^s = \sigma\left(\phi_g^s([\mathbf{R}_x^s; \mathbf{R}_y^s; \mathbf{D}^s])\right), \quad (14)$$

where $[\cdot; \cdot]$ denotes channel-wise concatenation, $\sigma(\cdot)$ is the sigmoid function, and $\mathbf{G}^s \in \mathbb{R}^{B \times C'_s \times H_s \times W_s}$ is the learned gating map. In practice, $\phi_g^s(\cdot)$ is implemented as a lightweight gating network consisting of a 1×1 convolution, GroupNorm, GELU, and a final 1×1 convolution. This design maps the concatenated compact features back to the original channel dimension, enabling channel-wise and spatially adaptive fusion over the full stage features.

The fused feature map is then formulated as

$$\tilde{\mathbf{F}}^s = \mathbf{G}^s \odot \tilde{\mathbf{X}}^s + (1 - \mathbf{G}^s) \odot \tilde{\mathbf{Y}}^s, \quad (15)$$

where the gating map controls the contribution of the two modalities at each spatial location and channel. Finally, the fused feature map is reshaped back into token form:

$$\mathbf{F}^s = \text{Reshape}(\tilde{\mathbf{F}}^s) \in \mathbb{R}^{B \times L_s \times C_s}. \quad (16)$$

In this way, DGFM performs lightweight yet effective multimodal fusion by explicitly modeling cross-modal discrepancy and using it to guide adaptive feature selection. In the overall framework, CPIA facilitates pre-stage token-level semantic interaction under the frozen backbone, whereas DGFM performs post-stage feature-level adaptive fusion on the stage outputs.

3.4. Modality-conditional random masking

Although the proposed dual-stream architecture enables cross-modal interaction, the optimization process may still become dominated by the modality that provides stronger immediate predictive cues, thereby weakening the contribution of the complementary modality. To mitigate this issue, we introduce a training-only strategy termed *Modality-Conditional Random Masking* (MCRM). The key idea is to simulate partial degradation in a modality-conditional manner: for a subset of training samples, only one modality is locally corrupted, while the counterpart remains intact. This design explicitly encourages the network to exploit complementary cross-modal evidence when one modality becomes unreliable.

Balanced modality-conditional masking. Given a mini-batch $\{(I_{opt}^{(i)}, I_{aux}^{(i)})\}_{i=1}^B$, we first select $N = \lfloor rB \rfloor$ samples for masking, where $r \in [0, 1]$ denotes the masking ratio. The selected samples are then partitioned into an optical-masked subset and an auxiliary-masked subset in a nearly balanced manner:

$$|B_{opt}| = \left\lfloor \frac{N}{2} \right\rfloor,$$

$$|B_{\text{aux}}| = N - |B_{\text{opt}}|, \\ B_{\text{full}} = \{1, \dots, B\} \setminus (B_{\text{opt}} \cup B_{\text{aux}}). \quad (17)$$

We define a modality-conditional masking operator $\mathcal{M}(\cdot)$ that masks K_m randomly sampled local rectangular regions on a single modality by setting the pixels within each region to zero. In all experiments, we set $K_m = 1$, i.e., each selected sample contains one masked rectangular block. The block area ratio is uniformly sampled from $[0.1, 0.3]$ of the input patch area, the aspect ratio is uniformly sampled from $[0.5, 2.0]$, and the block location is randomly sampled within the input patch. The corrupted inputs are formulated as

$$(\tilde{I}_{\text{opt}}^{(i)}, \tilde{I}_{\text{aux}}^{(i)}) = \begin{cases} (\mathcal{M}(I_{\text{opt}}^{(i)}), I_{\text{aux}}^{(i)}), & i \in B_{\text{opt}}, \\ (I_{\text{opt}}^{(i)}, \mathcal{M}(I_{\text{aux}}^{(i)})), & i \in B_{\text{aux}}, \\ (I_{\text{opt}}^{(i)}, I_{\text{aux}}^{(i)}), & i \in B_{\text{full}}. \end{cases} \quad (18)$$

Therefore, for each selected sample, only one modality is locally degraded while the other remains intact. Such asymmetric corruption prevents the model from over-relying on a single dominant modality and promotes the learning of complementary multimodal representations under partial modality degradation.

Training objective under MCRM. Given the MCRM-corrupted input pair $(\tilde{I}_{\text{opt}}, \tilde{I}_{\text{aux}})$, the network produces a fused prediction P together with two modality-specific auxiliary predictions, denoted by P_{opt} and P_{aux} :

$$(P, P_{\text{opt}}, P_{\text{aux}}) = F(\tilde{I}_{\text{opt}}, \tilde{I}_{\text{aux}}). \quad (19)$$

The fused prediction is supervised by the standard cross-entropy loss:

$$\mathcal{L}_{\text{main}} = \mathcal{L}_{\text{ce}}(P, Y_{\text{gt}}). \quad (20)$$

To explicitly guide the modality-specific branches to compensate for the corrupted modality, we further impose auxiliary supervision on pixels that remain misclassified by the fused branch. Specifically, we define a hard-pixel set based on the fused prediction as

$$\Omega = \{u \mid \arg \max_c P(u, c) \neq Y_{\text{gt}}(u)\}. \quad (21)$$

Only pixels in Ω are used to supervise the auxiliary branches, while correctly predicted pixels are ignored. The auxiliary loss is thus defined as

$$\mathcal{L}_{\text{aux}} = \mathcal{L}_{\text{ce}}(P_{\text{opt}}, Y_{\text{gt}}; \Omega) + \mathcal{L}_{\text{ce}}(P_{\text{aux}}, Y_{\text{gt}}; \Omega). \quad (22)$$

This hard-pixel-guided design avoids redundant supervision on already well-recognized regions and instead encourages each modality-specific branch to provide complementary cues precisely where multimodal fusion remains insufficient.

The overall training objective is formulated as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{main}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}}, \quad (23)$$

where λ_{aux} is the balancing coefficient for the auxiliary supervision and is set to 0.05 in all experiments.

In this way, the fused branch is optimized to learn robust multimodal representations, while the modality-specific auxiliary branches are explicitly driven to compensate for locally corrupted inputs on error-prone pixels. Since MCRM is used only during training, it introduces no additional inference-time computation. The increased parameters in the full model mainly come from the lightweight auxiliary heads for MCRM supervision, which are removed at test time.

4. Experiments

4.1. Datasets

We evaluate our method on two multimodal benchmarks: ISPRS Vaihingen and ISPRS Potsdam. Both datasets comprise high-resolution true orthophotos (TOP) and digital surface models (DSM), with pixel-wise annotations for six classes: impervious surfaces, building, low vegetation, tree, car, and clutter.

ISPRS vaihingen. The ISPRS Vaihingen dataset was acquired over the town of Vaihingen, Germany, covering a typical small urban area dominated by detached houses and multi-story residential buildings. It contains 33 TOP tiles with varying spatial extents at a ground sampling distance (GSD) of approximately 9 cm. The optical imagery includes three spectral channels: near-infrared (NIR), red (R), and green (G), and DSM is provided as the auxiliary modality. In our experiments, all three available TOP channels are used as the optical input, following the NIR-R-G order. Following the official ISPRS split, we use 16 tiles for training and the remaining 17 tiles for testing.

ISPRS potsdam. The ISPRS Potsdam dataset depicts a typical historical urban scene with dense building blocks, narrow streets, and complex urban textures. It has a higher spatial resolution with a GSD of approximately 5 cm and consists of 38 TOP tiles of a uniform size (6000×6000 pixels). The optical imagery contains four spectral channels (NIR, R, G, and B), and DSM is provided accordingly. In our experiments, we use the official RGB tiles as the three-band optical input, while the available NIR band is not used. Following the official split protocol, we use 24 tiles for training and the remaining 14 tiles for testing and evaluation.

4.2. Implementation details

All experiments are implemented in PyTorch and conducted on a single NVIDIA GeForce RTX 4090 GPU (24 GB memory). We employ pretrained DINOv2 and SAM as the feature extraction backbones for comparison. The models are optimized using AdamW with an initial learning rate of 3×10^{-4} , a weight decay of 0.01, and a batch size of 24. The learning rate schedule consists of a linear warm-up for the first 5 epochs followed by cosine annealing. During training, we apply random cropping with a crop size of 256×256 , together with random flipping and mirroring for data augmentation. Since training samples are generated on the fly from large ISPRS tiles, one epoch is defined as 1000 mini-batch iterations rather than a deterministic full pass over all possible crops. All models are trained for 50 such epochs. The checkpoint from the last training epoch is used for final testing, and the same checkpoint selection rule is applied to all re-implemented models and ablation variants. We adopt UPerNet (Xiao et al., 2018) as the segmentation decoder to produce pixel-wise prediction masks. The DSM auxiliary modality is independently min-max normalized to $[0, 1]$ for each tile during both training and testing. Optical images are first scaled to $[0, 1]$; ImageNet normalization is additionally applied only when the encoder belongs to the DINOv2 family, while no further ImageNet standardization is used for other encoders. The random seed is fixed to 42 for reproducibility. In MCRM, we use the same setting for both Vaihingen and Potsdam. The masking ratio is fixed to $r = 0.25$, so that 25% of each mini-batch is selected for modality-conditional local masking and partitioned as evenly as possible between optical-masked and auxiliary-masked samples. The detailed masking operator and auxiliary-loss weight are defined in Section 3.4. MCRM is enabled only during training and disabled during testing/inference.

We evaluate segmentation performance using Overall Accuracy (OA), mean F1-score (mF1), and mean Intersection-over-Union (mIoU). Following the standard evaluation protocol on ISPRS benchmarks, OA is computed as pixel-wise accuracy over all classes, including the background category, whereas mF1 and mIoU are averaged over the five foreground classes only. This setting enables a fair comparison with representative existing approaches.

4.3. Performance comparison

To evaluate the effectiveness and backbone compatibility of the proposed MoBaNet for multimodal remote sensing semantic segmentation, we benchmark it against representative baselines spanning different modeling paradigms, including classical CNN-based segmentation networks, strong Transformer-based segmenters, and multimodal

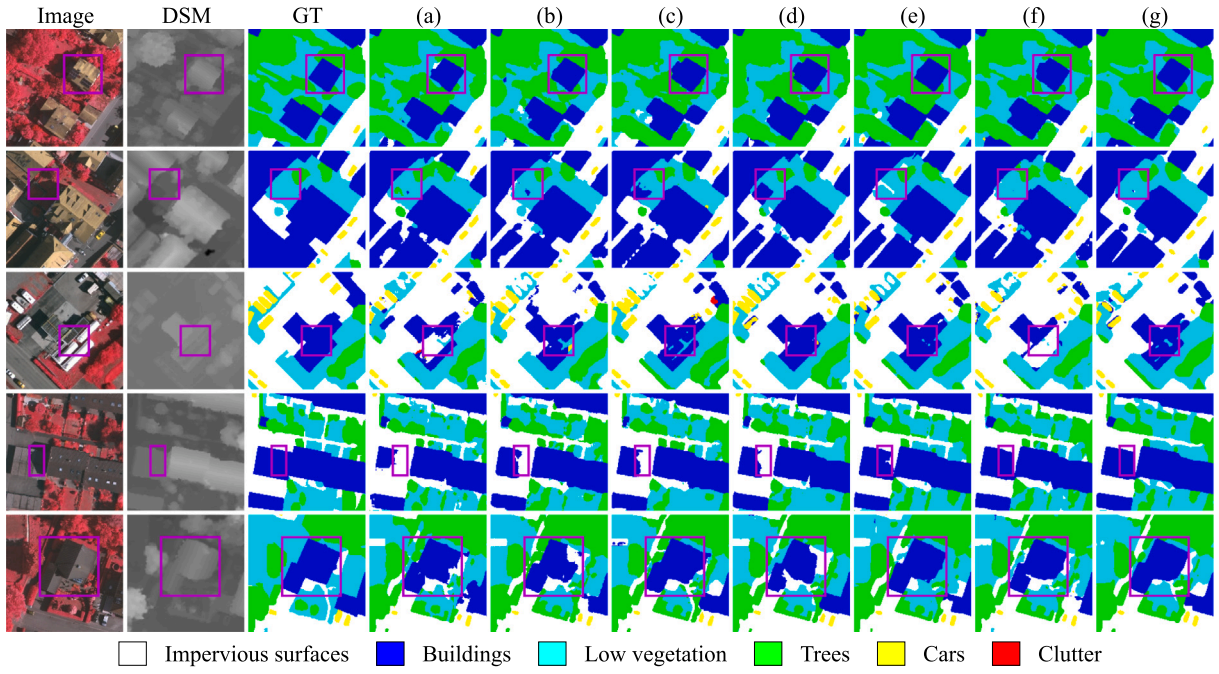


Fig. 4. Qualitative comparisons on 512×512 -pixel patches from the Vaihingen test set. (a) U-Net, (b) ABCNet, (c) DC-Swin, (d) UNetFormer, (e) FTransUNet, (f) MFNet, and (g) proposed MoBaNet. The samples are selected from representative challenging test scenes, and purple boxes highlight regions where prediction differences among methods are visually pronounced.

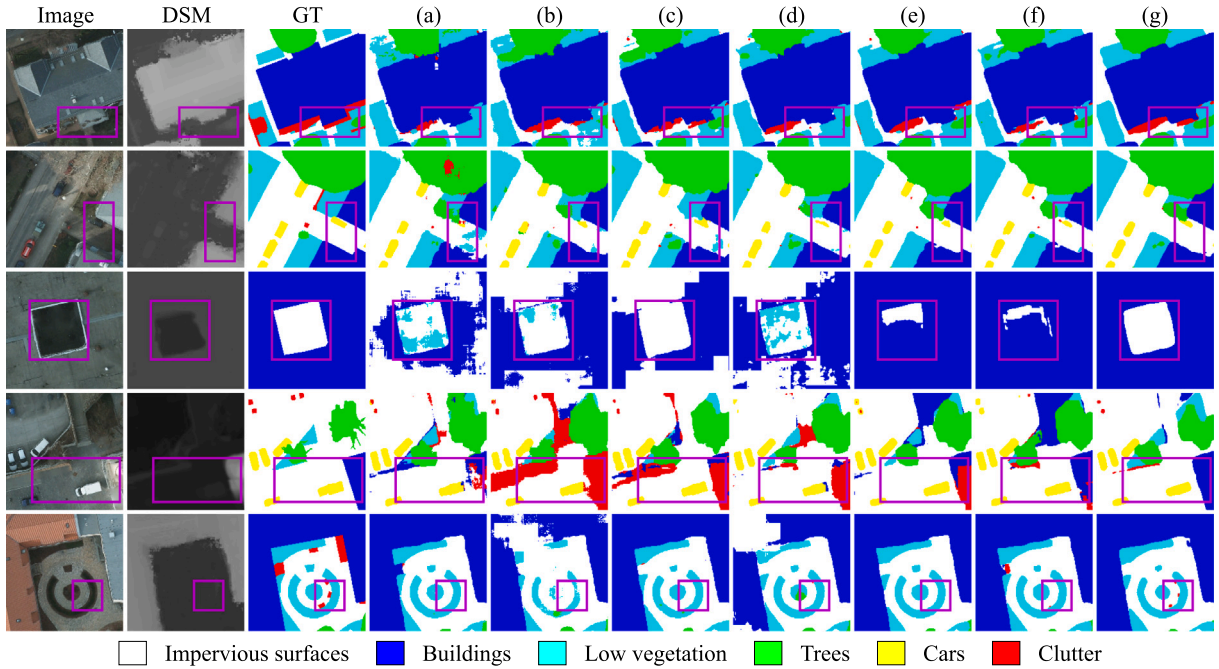


Fig. 5. Qualitative comparisons on 512×512 -pixel patches from the Potsdam test set. (a) U-Net, (b) ABCNet, (c) DC-Swin, (d) UNetFormer, (e) FTransUNet, (f) MFNet, and (g) proposed MoBaNet. The samples are selected from representative challenging test scenes, and purple boxes highlight regions where prediction differences among methods are visually pronounced.

frameworks equipped with explicit cross-modal fusion mechanisms. Specifically, UNet (Ronneberger et al., 2015), SegNet (Badrinarayanan et al., 2017), PSPNet (Zhao et al., 2017), ABCNet (Li et al., 2021), DC-Swin Wang et al. (2022a), MResU-Net (R. Li et al., 2022), and UNetFormer (Wang et al., 2022b) are adopted as strong optical-only baselines. These methods take only optical imagery as input, characterizing the attainable performance without geometric cues and serving

as a reference for quantifying the gains brought by multimodal fusion. In contrast, FuseNet (Hazirbas et al., 2017), vFuseNet (Audebert et al., 2018), CMFNet (Ma et al., 2022), CMGFNet (Hosseinpour et al., 2022), TMFNet (Liu et al., 2024a), FTransUNet (Ma et al., 2024c), and MFNet (Ma et al., 2025) further incorporate DSM on top of optical inputs and perform explicit fusion, representing typical designs for multimodal segmentation. To promote a fair comparison, we adopt

Table 1

Quantitative comparison on the Vaihingen dataset. Class-wise results are F1-scores; OA uses all classes, and mF1/mIoU use the five foreground classes. Best and second-best results are shown in bold and underlined, respectively.

Method	Backbone	Imp.	Bui.	Low.	Tre.	Car	OA	mF1 (%)	mIoU (%)
U-Net (Ronneberger et al., 2015)	–	86.63	90.56	76.84	83.75	72.80	80.73	82.12	70.17
SegNet (Badrinarayanan et al., 2017)	–	86.72	91.31	78.82	84.87	69.66	81.55	82.27	70.55
PSPNet (Zhao et al., 2017)	ResNet101	92.79	95.46	84.51	89.94	88.61	90.85	90.26	82.58
ABCNet (Li et al., 2021)	ResNet18	<u>96.67</u>	95.44	83.67	89.61	87.19	91.14	90.51	83.04
DC-Swin (Wang et al., 2022a)	Swin-B	96.68	95.85	84.38	89.59	86.31	91.33	90.56	83.13
MAResU-Net (R. Li et al., 2022)	ResNet18	91.97	95.04	83.74	89.35	78.28	90.05	87.68	80.75
UNetFormer (Wang et al., 2022b)	ResNet18	96.57	95.20	83.43	89.35	86.15	90.76	90.14	82.44
FuseNet (Hazirbas et al., 2017)	VGG16	91.66	96.28	78.98	90.28	81.37	90.51	87.71	78.71
vFuseNet (Audebert et al., 2018)	VGG16	91.0	94.4	84.5	89.9	86.3	90.0	89.22	–
CMFNet (Ma et al., 2022)	VGG16	92.36	97.17	80.37	90.82	85.47	91.40	89.48	81.44
CMGFNet (Hosseinpour et al., 2022)	ResNet34	92.81	96.20	84.08	90.26	85.02	90.98	89.67	81.60
TMFNet (Liu et al., 2024a)	ResNet34	91.25	95.70	80.87	88.82	84.60	89.71	88.25	78.70
FTransUNet (Ma et al., 2024c)	R50-ViT-B	92.89	96.02	84.70	89.80	86.94	91.05	90.07	82.18
MFNet (Ma et al., 2025)	ViT-B	92.79	96.14	84.15	90.57	86.78	91.35	90.33	82.60
MoBaNet (SAM)	ViT-B	93.21	96.51	85.72	91.58	90.06	91.65	90.97	83.66
	ViT-L	94.25	96.35	84.64	92.07	89.14	<u>91.80</u>	<u>91.29</u>	<u>84.17</u>
MoBaNet (DINOv2)	ViT-B	93.40	96.78	<u>85.54</u>	90.55	<u>89.52</u>	91.78	91.16	83.97
	ViT-L	94.18	<u>96.90</u>	84.58	<u>92.03</u>	88.32	91.90	91.39	84.37

Table 2

Quantitative comparison on the Potsdam dataset. Class-wise results are F1-scores; OA uses all classes, and mF1/mIoU use the five foreground classes. Best and second-best results are shown in bold and underlined, respectively.

Method	Backbone	Imp.	Bui.	Low.	Tre.	Car	OA	mF1 (%)	mIoU (%)
U-Net (Ronneberger et al., 2015)	–	86.62	91.96	81.52	84.09	83.55	81.93	85.55	74.92
SegNet (Badrinarayanan et al., 2017)	–	86.77	92.43	79.59	81.79	82.97	81.14	84.71	73.75
PSPNet (Zhao et al., 2017)	ResNet101	93.36	96.97	87.75	88.50	95.42	91.08	92.40	84.88
ABCNet (Li et al., 2021)	ResNet18	92.26	95.13	85.82	87.96	94.80	89.70	91.19	84.03
DC-Swin (Wang et al., 2022a)	Swin-B	93.65	96.31	87.16	88.31	95.27	90.86	92.14	85.64
MAResU-Net (R. Li et al., 2022)	ResNet18	91.4	95.6	85.8	86.6	93.3	89.0	90.5	83.9
UNetFormer (Wang et al., 2022b)	ResNet18	92.80	95.22	86.83	88.19	95.88	90.28	91.78	85.03
FuseNet (Hazirbas et al., 2017)	VGG16	92.64	97.48	87.31	85.14	96.10	90.58	91.60	84.86
vFuseNet (Audebert et al., 2018)	VGG16	92.7	96.3	87.3	88.5	95.4	90.6	92.0	–
CMFNet (Ma et al., 2022)	VGG16	92.84	97.63	88.00	87.40	95.68	91.16	92.10	85.63
CMGFNet (Hosseinpour et al., 2022)	ResNet34	92.62	97.20	86.46	88.55	94.46	90.59	91.86	85.18
TMFNet (Liu et al., 2024a)	ResNet34	92.83	96.20	89.86	89.09	91.80	91.05	92.21	85.71
FTransUNet (Ma et al., 2024c)	R50-ViT-B	92.70	97.24	87.53	88.64	94.39	90.78	92.10	85.56
MFNet (Ma et al., 2025)	ViT-B	93.94	97.46	89.72	86.41	95.17	90.50	91.82	85.08
MoBaNet (SAM)	ViT-B	<u>95.15</u>	96.64	90.84	87.45	94.83	91.27	92.58	86.38
	ViT-L	95.64	97.12	<u>90.78</u>	85.95	94.88	91.39	92.73	86.65
MoBaNet (DINOv2)	ViT-B	94.06	97.57	88.78	90.29	<u>96.29</u>	<u>92.11</u>	<u>93.40</u>	<u>87.80</u>
	ViT-L	94.16	<u>97.58</u>	89.08	<u>90.17</u>	96.66	92.23	93.53	88.03

consistent DSM co-registration/alignment, normalization, preprocessing, dataset splits, and evaluation metrics whenever possible, thereby reducing potential biases caused by inconsistent modality handling. Moreover, to examine the backbone-agnostic property of our framework, we implement and evaluate MoBaNet with SAM and DINOv2, demonstrating its stability and generalization across different backbones. Overall, the comparative quantitative results are reported in Tables 1 and 2.

Performance comparison on the vaihingen dataset. Table 1 summarizes the quantitative results on the ISPRS Vaihingen test set. MoBaNet achieves the best overall performance in terms of OA, mF1, and mIoU, demonstrating that the proposed frozen-backbone multimodal adaptation scheme can effectively exploit the complementarity between optical imagery and DSM while keeping the VFM backbone frozen. Specifically, MoBaNet (DINOv2-ViT-L) attains 91.90% OA, 91.39% mF1, and 84.37% mIoU, which are the best results in Table 1. Compared with the strong optical-only baseline DC-Swin, MoBaNet improves mIoU by 1.24 percentage points, indicating that DSM-provided geometric cues help reduce appearance-induced inter-class confusion and boundary deviations. Furthermore, MoBaNet outperforms the recent multimodal method MFNet (ViT-B) by 1.77 percentage points in mIoU (84.37% vs. 82.60%), suggesting that under a constrained

trainable-parameter budget, the proposed interaction-and-fusion design (CPIA/DGFM) together with the bias-mitigating training strategy (MCRM) can better activate discriminative cues from the auxiliary modality instead of converging to optical-dominant shortcut solutions.

Fig. 4 shows qualitative comparisons on the Vaihingen test set. MoBaNet (Fig. 4(g)) produces predictions that better preserve object integrity and yield sharper boundary delineation. In shadowed regions or areas with pronounced illumination variation, optical-only methods are more susceptible to spectral perturbations, often resulting in local misclassification, boundary bleeding, or fragmented masks. By contrast, MoBaNet generates more coherent segments with fewer artifacts for fine-grained and small-scale categories such as low vegetation and car, reflecting a more effective utilization of DSM structural and height cues through cross-modal interaction and discrepancy-guided adaptive fusion.

Performance comparison on the potsdam dataset. Table 2 reports the quantitative comparisons on the ISPRS Potsdam test set. MoBaNet again achieves the best performance across OA, mF1, and mIoU, indicating that the proposed framework remains effective in higher-resolution urban scenes with richer textures and more complex object layouts. In particular, MoBaNet (DINOv2-ViT-L) reaches 92.23% OA, 93.53% mF1, and 88.03% mIoU, which are the best results in Table 2. Compared with the strong optical-only baseline DC-Swin, MoBaNet improves mIoU by

Table 3

Comparison with representative generic PEFT baselines on the ISPRS Potsdam dataset using DINOv2-ViT-B. All methods use the same optical-DSM inputs, UPerNet decoder, input preprocessing, and training protocol. Train. Params denotes trainable parameters, and the percentage in parentheses denotes the ratio of trainable parameters to total model parameters.

Method	Adaptation strategy	Train. Params (M, %)	OA (%)	mF1 (%)	mIoU (%)	4mIoU
Frozen	Decoder-only adaptation	1.91 (2.16%)	90.47	91.74	84.95	–
VPT	Prompt tuning	1.92 (2.17%)	90.71	92.26	85.84	+0.89
Adapter	Bottleneck adapter	3.70 (4.10%)	91.65	92.91	86.94	+1.99
LoRA	Low-rank update	2.79 (3.13%)	91.64	93.05	87.20	+2.25
MoBaNet	CPIA + DGFM + MCRM	5.07 (5.43%)	92.11	93.40	87.80	+2.85

Table 4

Computational complexity on a 256×256 input using a single NVIDIA RTX 4090 GPU. MoBaNet uses DINOv2-ViT-B, and mIoU is reported on Potsdam. Bold values indicate the best results.

Method	Train. Params (M)	Latency (ms)↓	FPS↑	Memory (MB)↓	mIoU (%)↑
ABCNet	13.67	3.55	281.61	194.59	84.03
DC-Swin	118.39	14.59	68.56	603.83	85.64
UNetFormer	11.72	3.58	279.51	186.92	85.03
CMGFNet	43.68	13.13	76.18	357.94	85.18
FTransUNet	203.40	38.90	25.71	930.35	85.56
MFNet	20.42	105.36	9.49	1741.83	85.08
MoBaNet	5.07	13.97	71.60	402.99	87.80

Note: Train. Params denotes optimized parameters. For fully trainable baselines, it equals total parameters. MoBaNet has 93.44M total parameters, including the frozen DINOv2-ViT-B backbone, with 5.07M trainable parameters (5.43%). MCRM auxiliary heads are counted in Train. Params but removed during inference.

2.39 percentage points (88.03% vs. 85.64%), suggesting that DSM-derived geometric/height cues are especially beneficial for reducing confusion among visually similar categories such as impervious surfaces, building, and low vegetation in dense urban areas. Moreover, MoBaNet surpasses the recent multimodal baseline MFNet by 2.95 percentage points in mIoU (88.03% vs. 85.08%), implying that effective exploitation of auxiliary modalities requires not only a fusion head, but also targeted cross-modal interaction, discrepancy-aware feature selection, and training-time regularization under a constrained trainable-parameter budget. Notably, MoBaNet achieves strong performance on challenging categories such as low vegetation and car (e.g., 89.08% and 96.66%, respectively), further reflecting its improved discriminative capability for fine-grained targets and ambiguous regions.

Fig. 5 presents qualitative comparisons on the Potsdam test set. Owing to its higher spatial resolution and intricate urban textures, accurate segmentation requires preserving small objects, recovering regular geometric structures, and suppressing spurious predictions in cluttered backgrounds. As highlighted by the purple boxes, competing methods tend to miss or confuse small targets and produce noisy or broken masks in challenging regions, whereas MoBaNet yields cleaner predictions with more coherent regions and boundaries that better follow geometric contours. In addition, MoBaNet maintains clearer delineation at class transitions (e.g., road-building and vehicle-road interfaces), demonstrating improved robustness and fine-detail control in dense urban scenes.

4.4. Comparison with generic PEFT baselines

To further examine the effectiveness of MoBaNet's multimodal-specific adaptation design, we compare it with representative generic PEFT baselines on the ISPRS Potsdam dataset using DINOv2-ViT-B. This setting follows the default backbone used in the complexity and ablation studies, enabling a controlled comparison under the same optical-DSM inputs, UPerNet decoder, input preprocessing, training schedule, and checkpoint selection rule.

We consider three generic PEFT baselines: VPT, Adapter, and LoRA. For VPT, learnable prompt tokens are inserted into the input token

space of each modality stream. For Adapter, modality-specific bottleneck adapters are inserted into both the optical and DSM streams. For LoRA, low-rank updates are added to the query and value projections of the self-attention layers in both streams. In all generic PEFT baselines, the DINOv2 backbone is kept frozen, and no cross-modal prompt generation, discrepancy-guided fusion, or MCRM training is introduced.

As shown in Table 3, all generic PEFT baselines clearly improve over the frozen backbone base, demonstrating that lightweight adaptation is beneficial for transferring frozen VFMs to optical-DSM segmentation. Among them, LoRA achieves the strongest generic-PEFT result with 87.20% mIoU, followed by Adapter with 86.94% and VPT with 85.84%. MoBaNet further improves the mIoU to 87.80%, outperforming LoRA, Adapter, and VPT by 0.60, 0.86, and 1.96 percentage points, respectively. These results indicate that generic stream-wise PEFT can enhance frozen-backbone adaptation, but explicitly allocating trainable capacity to cross-modal prompt interaction, discrepancy-guided fusion, and modality-robust training leads to more effective multimodal adaptation.

4.5. Model complexity analysis

To evaluate computational complexity and efficiency, we compare MoBaNet with representative optical-only and multimodal baselines in terms of trainable parameters, inference latency, FPS, peak memory footprint, and mIoU, as reported in Table 4. The MoBaNet configuration uses the DINOv2-ViT-B backbone. These metrics reflect the trade-off between segmentation performance and deployment cost.

MoBaNet achieves the highest mIoU (87.80%) with only 5.07M trainable parameters, the smallest trainable-parameter count in the table. The “Train. Params (M)” column reports the number of optimized parameters. For fully trainable baselines, trainable parameters are equal to total parameters. In contrast, MoBaNet keeps the DINOv2-ViT-B backbone frozen and optimizes only the lightweight multimodal adaptation modules, auxiliary patch embedding, decoder head, and training-time auxiliary heads used by MCRM. Therefore, the frozen backbone is excluded from the trainable-parameter count but included in the total parameter count reported in the table note. Specifically, MoBaNet contains 93.44M total parameters, of which only 5.07M are trainable, corresponding to a trainable-parameter ratio of 5.43%. Compared with multimodal baselines, MoBaNet is lighter than CMGFNet (43.68M), FTransUNet (203.40M), and MFNet (20.42M) in terms of trainable parameters, while improving mIoU; it exceeds FTransUNet and MFNet by 2.24 and 2.72 percentage points, respectively. These results support MoBaNet's parameter allocation strategy: instead of updating the full foundation backbone, the trainable budget is concentrated on lightweight modules for multimodal adaptation and fusion.

From the perspective of deployment efficiency, MoBaNet also exhibits a favorable balance between performance and hardware cost. Its inference latency is 13.97 ms with a corresponding FPS of 71.60, which is close to that of the efficient multimodal baseline CMGFNet (13.13 ms, 76.18 FPS), while being markedly better than FTransUNet and MFNet in both speed and memory usage. Although lightweight optical-only methods such as ABCNet and UNetFormer remain faster and more memory-efficient, they do not exploit complementary DSM information

Table 5

Ablation study of the proposed key designs on the ISPRS Potsdam dataset. “Base” denotes the frozen VFM dual-stream baseline with a trainable auxiliary patch embedding and the same decoder head, without any of the proposed designs.

Variant	CPIA	DGFM	MCRM	Train. Params (M)	OA (%)	mF1 (%)	mIoU (%)
Base				1.91	90.47	91.74	84.95
Base + CPIA	✓			3.99	91.43	92.57	86.33
Base + CPIA + DGFM	✓	✓		5.00	91.73	93.02	87.15
Base + CPIA + DGFM + MCRM (Full)	✓	✓	✓	5.07	92.11	93.40	87.80

Table 6

Modality-missing evaluation at inference time on Potsdam. Here RGB denotes the selected optical input used for Potsdam, and “RGB-only” sets DSM to zeros. All models are trained with full RGB+DSM inputs unless otherwise specified. “Drop (↓)” is computed as the performance difference between RGB+DSM and RGB-only.

Dataset	Input	MoBaNet (Full)			Reference baseline		
		OA (%)	mF1 (%)	mIoU (%)	OA (%)	mF1 (%)	mIoU (%)
Potsdam	RGB + DSM	92.11	93.40	87.80	91.73	93.02	87.15
	RGB-only	91.36	92.74	86.63	72.38	76.97	64.20
	Drop (↓)	0.75	0.66	1.17	19.35	16.05	22.95

and therefore achieve lower segmentation accuracy than MoBaNet. Overall, these results indicate that MoBaNet provides a strong trade-off among segmentation accuracy, parameter efficiency, and inference cost, highlighting its practical potential for multimodal remote sensing segmentation under resource-constrained deployment settings.

4.6. Ablation study

To evaluate the contribution of each proposed component and the robustness of the overall framework, we conduct two groups of ablation experiments under the same training protocol described in Section 4, using DINOv2-ViT-B as the default backbone. Unless otherwise specified, all variants are evaluated in terms of OA, mF1, and mIoU, and the number of trainable parameters is additionally reported to reflect parameter efficiency.

Component ablation. We first investigate the contribution of each module by progressively introducing CPIA, DGFM, and MCRM on top of a frozen dual-stream baseline, denoted as *Base*. Specifically, *Base* consists of the frozen backbone with a trainable auxiliary patch embedding and the same decoder head, but without any of the proposed components. Starting from *Base*, incorporating **CPIA** yields consistent improvements across all evaluation metrics, indicating that prompt-injected adapters effectively facilitate cross-modal semantic interaction under a frozen backbone. Further introducing **DGFM** brings additional gains, suggesting that discrepancy-guided adaptive fusion is beneficial for suppressing noisy responses and improving dense prediction quality. Finally, adding **MCRM** leads to the best overall performance, confirming that Modality-Conditional Random Masking helps mitigate optical-dominant optimization and promotes a more balanced exploitation of complementary DSM cues. Overall, the full model achieves the best trade-off between segmentation accuracy and parameter efficiency, demonstrating the effectiveness of the proposed multimodal adaptation design. It is worth noting that the *Base* variant already uses a frozen dual-stream VFM and the same decoder; its lower performance indicates that simply freezing a VFM and attaching a decoder is insufficient, while the consistent gains from CPIA, DGFM, and MCRM confirm the necessity of allocating trainable capacity to multimodal-specific interaction, fusion, and robustness regularization.

Auxiliary-modality-missing robustness. We further evaluate the robustness of the model under modality-missing inference to examine whether the learned representation is overly dependent on a single modality. The comparison is conducted on the Potsdam dataset. Specifically, two inference settings are considered: *RGB+DSM* and *RGB-only*, where the latter is obtained by setting DSM to zeros while keeping all network

parameters unchanged. To isolate the effect of the proposed regularization, the reference model is the variant trained without MCRM, i.e., *Base+CPIA+DGFM*. As shown in Table 6, the full model exhibits substantially smaller performance degradation when DSM is removed. In particular, under the *RGB-only* setting, the performance drop of MoBaNet is 0.75% in OA, 0.66% in mF1, and 1.17% in mIoU, whereas the reference baseline suffers much larger drops of 19.35%, 16.05%, and 22.95%, respectively. These results indicate that MCRM effectively reduces the model’s over-reliance on a single dominant modality and improves the robustness of multimodal representation learning under partial modality degradation. Notably, MCRM is applied only during training and introduces no additional computational overhead at inference time.

Qualitative visualization. To complement the quantitative analysis, Fig. 6 presents class-specific response maps for the *Buildings* category on representative samples from the Potsdam dataset. Compared with *Base*, the full model produces more compact, spatially coherent, and semantically concentrated activations over building regions, while suppressing irrelevant responses in the surrounding background. In particular, the highlighted regions generated by the full model align more closely with the spatial extent and boundaries of buildings under varying roof appearances and complex local contexts. These qualitative observations are consistent with the quantitative improvements reported in Table 5 and provide further visual evidence that the proposed design enhances multimodal representation quality.

5. Discussion and future work

We discuss MoBaNet in relation to recent PEFT, multimodal fusion, and modality-robust learning designs, and further interpret the empirical findings with recent EO literature.

- (1) *Relation to generic PEFT and foundation-model adaptation:* Generic PEFT methods, such as Adapter (Houlsby et al., 2019), LoRA (Hu et al., 2022), and VPT (Jia et al., 2022), mainly adapt frozen backbones through stream-wise parameter insertion. Recent RS studies have also explored adapter-style or efficient tuning strategies for adapting foundation models to remote sensing tasks (Hu et al., 2024b,a; Zhang et al., 2025). As confirmed by our PEFT comparison, these methods improve the frozen backbone but remain less effective than MoBaNet. CPIA differs from them by generating prompts from paired modalities and injecting them into both modality streams, thereby coupling parameter-efficient adaptation with deep cross-modal interaction. These results suggest that allocating trainable parameters to multimodal-specific interaction is more effective than simply adding generic PEFT modules.

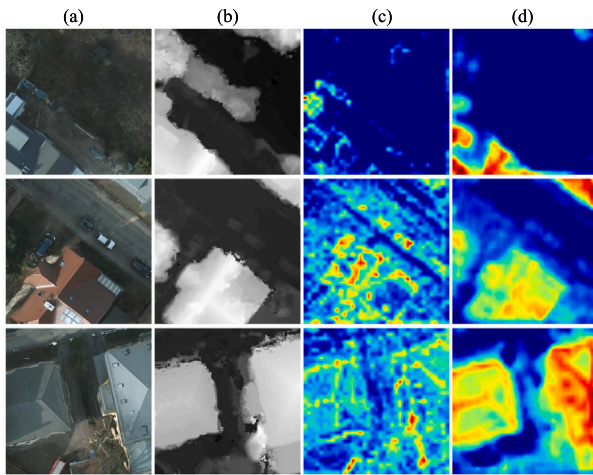


Fig. 6. Qualitative comparison of feature response visualizations on representative Potsdam samples for the *Buildings* category. (a) RGB image. (b) DSM. (c) Base. (d) Full model.

- (2) *Relation to gated and attention-based fusion:* Recent multimodal RS segmentation methods have explored Transformer-based, attention-based, gated, and deep fusion mechanisms (Ma et al., 2024c; Liu et al., 2024a; Ma et al., 2025; Wei et al., 2024; Liu et al., 2025). These studies show that effective fusion should selectively enhance complementary cues while suppressing redundant or noisy responses. DGM follows this selective-fusion principle but differs from conventional gated fusion by explicitly constructing a cross-modal discrepancy cue before predicting the adaptive gate. This design provides a plausible explanation for the observed gains over multimodal baselines to discrepancy-aware feature selection, especially in ambiguous urban regions where optical appearance and DSM geometry provide complementary evidence.
- (3) *Relation to modality-dropout-style robustness training:* MCRM is related to modality dropout, random masking, and incomplete-modality learning because it simulates partial modality degradation during training (Chen et al., 2024; Liu et al., 2025). However, it is not a simple modality-level dropout strategy. It locally corrupts only one modality at a time while keeping the counterpart modality intact, and further applies hard-pixel auxiliary supervision to modality-specific branches. This design encourages the model to exploit complementary cues where the fused prediction remains unreliable, which is consistent with the reduced degradation observed in the modality-missing evaluation.
- (4) *Limitations and future work:* Although the results on Vaihingen and Potsdam support the effectiveness of MoBaNet, several issues remain open. First, more systematic interpretability analyses, such as gate-weight statistics, modality-contribution measurement, or gradient-based attribution, are needed to better reveal how different modalities affect the final prediction. Second, MoBaNet is currently evaluated on two urban multimodal benchmarks, and future work should further examine its scalability to broader EO settings, such as optical-SAR, multispectral-LiDAR, or hyperspectral-LiDAR fusion. Recent multimodal RS foundation models and SAM-based adaptations, such as RingMoSAM (Yan et al., 2023), SkySense (Guo et al., 2024), and MM-SAM (Xue et al., 2025), may provide useful pretraining resources and adaptation paradigms for this direction.

6. Conclusion

In this paper, we proposed MoBaNet, a parameter-efficient multimodal adaptation framework for remote sensing semantic segmentation. Built upon a frozen vision foundation model, MoBaNet combines CPIA, DGM, and MCRM to enable cross-modal prompt interaction, discrepancy-guided adaptive fusion, and training-time robustness regularization.

Experiments on the ISPRS Vaihingen and Potsdam benchmarks show that MoBaNet performs favorably against the compared methods while requiring only 5.43% trainable parameters in the default DINOv2-ViT-B setting. The ablation and auxiliary-modality missing analyses further verified that the proposed modules contribute complementary benefits in semantic interaction, reliable fusion, and robustness under partial modality degradation. These results suggest that MoBaNet provides an effective and efficient solution for multimodal adaptation of large vision backbones in remote sensing segmentation.

CRedit authorship contribution statement

Haocheng Li: Writing – original draft, Software, Methodology, Investigation. **Juepeng Zheng:** Writing – review & editing, Methodology, Funding acquisition. **Shuangxi Miao:** Investigation, Funding acquisition, Conceptualization. **Ruibo Lu:** Validation, Resources, Conceptualization. **Guosheng Cai:** Resources, Project administration, Funding acquisition. **Haohuan Fu:** Writing – review & editing, Investigation, Funding acquisition. **Jianxi Huang:** Writing – review & editing, Funding acquisition, Formal analysis, Conceptualization.

Code availability statement

The source code of MoBaNet, including training and evaluation scripts as well as configuration files for reproducing the reported results, is publicly available at <https://github.com/sauryeo/MoBaNet>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Research Link Provided

[ISPRS Dataset \(Original data\)](#) (ISPRS Dataset)

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